



Antiviral activity of HuaganJiedu decoction (HGJDD) against hepatitis B virus (HBV) through FOXO4/ERK/HNF4 α signal pathway

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ABSTRACT

Ethnopharmacological relevance: Chronic hepatitis B virus (HBV) infection is still a widespread global health issue. HuaganJiedu Decoction (HGJDD) is a common prescription for treating HBV in China, which has the effect of enhancing antiviral efficacy and improving clinical efficacy. However, its precise mechanism of action remains unclear, warranting further investigation to elucidate its therapeutic potential and integration into standard medical practices.

Aim of the study: This study aims to explore the therapeutic mechanism of HuaganJiedu Decoction (HGJDD) in HBV.

Materials and methods: We investigated the therapeutic potential of HGJDD, and LC-MS analysis characterized the chemical profile of HGJDD. In vitro, we utilized HepG2.2.15 cell line to assess cytotoxicity and treatment efficacy of HGJDD compared to Entecavir controls. In vivo, assessments included monitoring HBV-related biomarkers and viral load. Network pharmacology and RNA-seq analyses identified molecular pathways and targets influenced by HGJDD treatment. Immunofluorescence and Western blotting provided further insights into the therapeutic mechanisms underlying HGJDD for HBV.

Results: HGJDD showed no toxicity on HepG2.2.15 cells at 10%, 20%, 40%, and 80% serum concentrations. In vitro, HGJDD reduced HBsAg, HBeAg, and HBV DNA levels by dose-dependently and time-dependently. HGJDD can decrease the levels of HBsAg, HBeAg, and HBV DNA in serum and liver levels, meanwhile the therapeutic effect of high-dose HGJDD approach to EVT's in HBV Tg mice. According to intersection of network pharmacology and transcriptome, FOXO signal pathway was highlighted as potential targets and Immunofluorescence find that FOXO4D protein expression lever was increased in three HGJDD group, especially in high-dose HGJDD group. Western blotting confirmed increased level of FOXO4, ERK, and p-ERK and decreased levels of HNF4 α , which reflected that the therapeutic effect was closely to FOXO4/ERK/HNF4 α signal pathway.

Conclusions: Traditional Chinese medicine (TCM) offers diverse herbal treatments for HBV, with HGJDD showing efficacy in reducing HBsAg, HBeAg, and HBV DNA levels at cellular and animal levels. This study identified that FOXO4/ERK/HNF4 α signal pathway played an important role in HGJDD's therapeutic effects. These findings support HGJDD's potential in HBV treatment, providing a scientific basis for clinical use.

1. Introduction

Hepatitis B virus (HBV) infection is acknowledged as the most

prevalent chronic viral disease around the world (Trépo et al., 2014), it still one of the major public health problem worldwide due to incomplete vaccine coverage, ongoing transmission, and a reservoir of persons

Abbreviations: HuaganJiedu Decoction, HGJDD; Traditional Chinese medicine, TCM; Hepatitis B virus, HBV; Hepatocellular carcinoma, HCC; Hepatitis B surface antigen, HBsAg; Hepatitis B e antigen, HBeAg; chronic hepatitis B, CHB; Entecavir, EVT; enzyme-linked immunosorbent assay, ELISA; Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform, TCMSP.

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infected prior to the availability of vaccination (Laupèze et al., 2023). Around 30% of the world’s population shows serological markers that suggest either ongoing or past Hepatitis B virus (HBV) infection (Hatzakis et al., 2013). Chronic hepatitis B is estimated to affect between 275 and 316 million individuals globally each year, resulting in approximately 820,000 fatalities and 1.5 million new infections (Kudaravalli et al., 2024). As a hepatotropic DNA virus, HBV is capable of establishing persistent chronic infections, which can lead to a range of liver-related complications, including dysfunction, fibrosis, cirrhosis, liver failure, and hepatocellular carcinoma (HCC) (Yuen et al., 2018). Chronic hepatitis B was linked with an estimated 360,000 cases of HCC globally in 2018 (de Martel et al., 2020). The occurrence rate of Hepatitis B virus (HBV) infection, as judged by the existence of HBsAg, reveals significant regional variations (Terrault et al., 2018). Although preventive measures for infants, childhood vaccination initiatives can decline in incidence rates, only 42% of infants received a dose of HBV vaccine within 24 h of birth globally (Laupèze et al., 2023). In 2021. Antiviral treatments have contributed to a decline in mortality, HBV remains a persistent issue due to the lack of curative virological therapies (Fanning et al., 2019).

The primary objective of contemporary therapeutic approaches is to diminish the occurrence of severe complications, including hepatocellular carcinoma (HCC), while enhancing the quality of life by mitigating liver inflammation and decreasing the likelihood of disease progression. The loss of hepatitis B surface antigen (HBsAg) is a critical factor in attaining the aim of reducing the incidence of HCC. The goal of current treatment strategies is to reduce the incidence of severe complications such as HCC and improve quality of life by reducing inflammation in the liver and lowering the risk of disease progression. Loss of HBsAg is the main contributor to achieving the goal of reducing HCC incidence (Lebossé et al., 2017). Nucleoside analogues (NAs), one of the most recommended oral antiviral treatment drugs, has been shown to be effective in inhibiting HBV replication, easing liver inflammation and fibrosis, preventing decompensation, and lowering the risk of HCC occurrence (Huang et al., 2023; Kudaravalli et al., 2024; Lok et al., 2017). Nonetheless, challenges such as inadequate evaluation, suboptimal treatment rates, and limited patient benefits persist despite the availability of NAs (Chen et al., 2018; Nguyen et al., 2019).

As a classical medical system with a history exceeding a millennium, Traditional Chinese medicine (TCM) has long been a vital source of inspiration for the development of novel therapeutic agents (Kaixian, 2020). TCM offers a unique perspective and approach to treating diseases that is deeply rooted in centuries of clinical experience and theoretical knowledge. Unlike Western medical approaches that predominantly rely on vaccines and antiviral drugs, TCM occupies a distinctive role within Chinese clinical practice in treatment of virus associated diseases. Herbal formulations grounded in TCM principles are integral to the management of HBV infection, with approximately 80% of chronic HBV patients in China receiving treatment through Chinese herbal medicine (Yin et al., 2022; Zhang et al., 2010). This widespread use reflects the trust and recognition of TCM in addressing liver diseases. TCM treatments are characterized by their multifactorial and multi-targeted properties. For example, many TCM herbs and formulas can simultaneously modulate the immune system, regulate liver function, and inhibit viral replication. This comprehensive approach may provide unique advantages in addressing the complexities of HBV infections (Cui et al., 2010; Xu et al., 2021; Zhou et al., 2023). The active constituents of various TCM therapies have been shown to facilitate HBV clearance, thereby serving as a significant adjunct to conventional antiviral treatments (Chen and Zhu, 2013).

In recent years, studies on specific TCM formulas have further demonstrated their potential in HBV treatment. HuaganJiedu Decoction (HGJDD) is an empirical formulation utilized in China for the management of CHB (Liu et al., 2010). Clinically, CHB is characterized by symptoms including abdominal distension, a bitter taste in the mouth, jaundice, yellow urine, and a thick, greasy tongue coating, which are

indicative of “dampness-heat accumulation” as understood in traditional Chinese medicine. The components of HGJDD predominantly possess properties that resolve dampness and detoxify, while also promoting blood circulation and enhancing overall vitality (Liu et al., 2010). The decoction comprises nine botanical ingredients, detailed in Table 1, and their nomenclature has been verified through World Flora Online (<http://www.worldfloraonline.org>) and the MPNS (<http://mpns.kew.org>). Numerous studies have demonstrated that HGJDD can enhance the efficacy of antiviral medications in the treatment of CHB, alleviate clinical symptoms, and decrease levels of HBsAg, HBeAg, and HBV-DNA (Chen, 2006; Li et al., 2018b; Sun and Mao, 2019; Yuan et al., 2022). Despite the affirmation from various clinical investigations regarding the significant improvement in antiviral effects when HGJDD is administered in conjunction with antiviral agents, as well as its role in facilitating the negative conversion of antigens, the specific effects of HGJDD as a standalone treatment for hepatitis B and the precise mechanisms underlying its action remain inadequately understood. This study aims to employ methodologies to investigate the therapeutic efficacy of HuaganJiedu Decoction at both cellular and animal levels, thereby elucidating its mechanisms of action in the treatment of HBV.

2. Methods

2.1. Chemicals and materials

The Chinese herb was obtained from Guang ‘anmen Hospital and stored in the Traditional Chinese Medicine Center for Prevention and Treatment of Viral Diseases in Institute of Basic Theory for Chinese Medicine. The used media for cell culture contained Dimethyl sulfoxide (Sigma, DH105-2, USA), penicillin streptomycin mixture (Biorigin, BN28050, China), G418 (Biorigin, BN20416, China), Entecavir (EVT, Yuanye Bio-Technology, B26389, China), MEM medium (INVITROGEN, C11095500BT, USA), fetal bovine serum (ExCell Bio, FSD500, China), CCK-8 (Dojindo, CK04, Japan). HBsAg and HBeAg kits were purchased from UpingBio technology Co., Ltd. (Hangzhou, Zhejiang, China). HBV DNA kit was purchased from Sansure Biotech Inc. (Changsha, Hunan, China). DNA RNA Protein Kit (300T) was purchased from OMEGA BIO-TEK (OMEGA, R6734-02, USA).

2.2. Composition of traditional Chinese Medicine, decocting methods, and extraction of medicinal serum

The varieties and ingredients of TCM in HuaganJiedu decoction were included 15 g of Polygonum cuspidatum, 15 g of Japanese Ardisia, 15 g of Scutellaria barbata, 20 g of Glabrous Greenbrier, 20 g of Stringy stonecrop, 10 g of Red peony root, 10 g of Turmeric, 10 g of Black soya bean and 3 g of Liquorice root, which were prepared according to the conventional decocting method, that is, traditional Chinese medicine was soaked in distilled water for 1 h and then decocted. Bring to the boil and cook for 30min. Finally, the volume was determined to 250ml/pair of Chinese medicine, which was used to extract drug-containing serum from rats and stored at 4 °C for later use. This is the daily dose of normal people, but also the basis for later mice or rats to take the dose, they are

Table 1
The ingredients of HGJDD.

English name	Botanical nomenclature	Chinese name
Polygonum Cuspidatum	<i>Reynoutria japonica</i> Houtt.	HuZhang
Japanese Ardisia	<i>Ardisia japonica</i> (Thunb.) Blume	AiDiCha
Scutellaria Barbata	<i>Scutellaria barbata</i> D. Don.	BanBianLian
Glabrous Greenbrier Rhizome	<i>Smilax glabra</i> Roxb.	TuFuLin
Stringy Stonecrop	<i>Sedum sarmentosum</i> Bunge	CuiPengCao
Red Peony Root	<i>Paeonia lactiflora</i> Pall.	ChiShao
Turmeri	<i>Curcuma longa</i> L.	JiangHuang
Black Soybean	<i>Glycine max</i> (L.) Merr.	HeiDou
Licorice	<i>Glycyrrhiza glabra</i> L.	GanCao

converted according to the body surface area.

A total of 20 3-month-old healthy SPF Wistar rats were randomly assigned to a blank control group and a Chinese medicine group (10 rats each), and given distilled water and Chinese medicine decoction, respectively. The drug was given intragastric at the rate of 22.5 mL/kg once a day for 7 days. One hour after the final administration, blood was collected from the abdominal aorta under aseptic conditions, left at 4 °C for 1 h, centrifuged for 15 min at 3000×g, and drug-containing serum was separated. The drug-containing serum of rats in the same group was mixed well, inactivated by water bath at 56 °C for 30 min, filtered by microporous filter membrane for bacteria removal, frozen tube packaging, and stored at −80 °C. Blank control serum will be used to dissolve EVT and dilute the drug-containing serum.

2.3. LC-MS analysis of HGJDD

The binary gradient elution system was composed of (A) water (containing 0.1% formic acid by volume) and (B) acetonitrile (containing 0.1% formic acid by volume). Separation was accomplished by using the gradient. Data acquisition was carried out in the full scan mode (with a mass-to-charge ratio range from 100 to 1000) combined with the IDA mode. The LC-MS data were processed by Progenesis QI V2.3 (Nonlinear Dynamics, Newcastle, UK) for baseline filtering, peak identification, integration, retention time correction, peak alignment, normalization, and analysis.

2.4. Cell culture

The human HBV-transfected cell line HepG2.2.15 (acquired from the China Center for Type Culture Collection, CCTCC, of Wuhan University, China) was cultivated in MEM medium supplemented with 10% fetal bovine serum (FBS), 100 units/ml penicillin, 100 µg/ml streptomycin, and 380 µg/ml G418 at 37 °C in a humidified atmosphere with 5% carbon dioxide. The cells were kept in culture plates in the medium at a density of 2×10^5 cells/ml and incubated for 24 h before being incubated with the indicated concentrations of the drugs. The medium was changed every three days.

2.5. Cytotoxicity assay

The cytotoxicity of drug serum was determined by the CCK-8 assay to identify the non-toxic concentrations of drug serum to the cultured cells (Wang et al., 2009). In brief, HepG2.2.15 cells were cultured in 96-well plates at a density of 1×10^4 cells per well and incubated for 24 h. Subsequently, drug serum of different concentrations (0%, 10%, 20%, 40%, 80%) and normal culture (MEM complete medium containing 10% FBS) was added to the wells for 3, 6, and 9 days. After the treatment, 10 µl of CCK-8 was added to each well and incubated for 2 h at 37 °C. The absorbance at a wavelength of 450 nm was measured using an automatic plate reader (Molecular Devices, SpectraMax M5, USA). The cell growth inhibitory rate was calculated using the following formula: the percent of cell viability (%) = [A450 of each group/A450 of 0% drug serum] × 100%. Concentrations were considered non-toxic if the corresponding cell viability was greater than 95% and there was no statistical difference compared with 0% drug serum. Each experiment was performed in triplicate, and each group contained 10 repetitive wells.

2.6. The treatment of HGJDD in vitro

The HepG2.2.15 cells were subjected to treatment with HGJDD serum at concentrations of 20%, 40%, and 80%, or with EVT (10 µmol/ml) for 3, 6, and 9 days (Wang et al., 2020). Negative control group were treated with serum without any ingredients of HGJDD (0% HGJDD serum). EVT was dissolved with no-drug serum as the positive control. Media was changed every 3 days during the treatment. The supernatants and cells were collected for the subsequent disposal and detection.

Cellular DNA, RNA, and Protein were extracted follows the protocol of OMEGA DNA RNA protein Kit.

2.7. The treatment of HGJDD in vivo

First, HGJDD was further concentrated to 50ml/para-Chinese medicine, and was used as mother liquor of Chinese medicine for mouse experiment, and stored at 4 °C for later use. The decoction was further diluted with distilled water into 1/2 and 1/4 times the original concentration as HGJDD medium dose and HGJDD low dose, and the original decoction was used as HGJDD high dose. Each mouse in each group was given 13 ml/kg body weight.

HBV Tg transgenic mice were used in this experiment. There are a total of five groups, namely the control group, high-dose HGJDD group (HGJDD-H), medium-dose HGJDD group (HGJDD-M), low-dose HGJDD group (HGJDD-L), and positive drug group (entecavir, EVT). The drugs are given by gavage once a day for a total of 6 weeks of intervention. We take blood from the eyeballs every two weeks before the intervention, and liver samples after 6 weeks of intervention. Liver DNA, RNA, and Protein were extracted follows the protocol of OMEGA DNA RNA protein Kit. These two samples are used to detect HBsAg, HBeAg, and HBV DNA. This work's animal studies were approved (Approval No. VST-PREPA-SOP-19-01) by the Animal Care Committee of Beijing Vitalstar Biotechnology Co., Ltd., who have qualifications to conduct animal-related experiments for hepatitis B.

2.8. The level of HBsAg and HBeAg

The levels of HBsAg and HBeAg in cell supernatants, cell lysates, serum, and liver lysates were measured using commercially available enzyme-linked immunosorbent assay (ELISA) kits in accordance with the manufacturer's protocol. The absorbance at 450 nm was determined by a microplate reader (BIO-RAD, 168-1000XC, USA). The specific concentration is determined according to the standard curve drawn.

2.9. The level of HBV DNA

The total DNA in cells and culture medium was extracted according to the protocol of the OMEGA DNA RNA Protein Kit (OMEGA BIO-TEK, R6734-02, NORCROSS, Georgia, USA). Hepatitis B Viral DNA quantitative Fluorescence Diagnostic Kit was used to detect HBV DNA level (Sansure Biotech Inc, China). A mixture of reaction reagents was put into a 0.2 ml reaction tube for PCR amplification using the ABI 7300 Real-Time PCR System (Applied Biosystems, Foster City, CA, USA). The amplification started with a UNG enzyme reaction at 50 °C for 2 min, followed by a step at 94 °C for 5 min. Then, there were 45 cycles consisting of denaturing at 94 °C for 15 s and annealing elongation at 57 °C for 30 s. The relative HBV DNA level was computed according to the standard curve.

2.10. Network pharmacology analysis for HGJDD

The chemical composition in HGJDD were retrieved from the Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform (TCMSP) (Ru et al., 2014). HBV targets were collected from the OMIM database (<https://omim.org>) and GeneCards database (<https://www.genecards.org>) (Safran et al., 2010). The targets of HGJDD and HBV were introduced into jvenn (<https://jvenn.toulouse.inra.fr/>) to identify the overlapping targets. Based on the common targets, pathways enrichment analysis was performed by R software (version 4.3.2) and drew by Bioinformatics online tool (www.bioinformatics.com.cn).

2.11. RNA-seq and functional enrichment analyses

Total RNA was extracted from three biological replicates per group

using the OMEGA DNA/RNA/Protein Kit (OMEGA BIO-TEK, R6734-02, Norcross, Georgia, USA). The quality and quantity of total RNA were assessed by an UltrasecTM 2100 pro UV/Visible Spectrophotometer (Amersham Biosciences, Uppsala, Sweden) and agarose gel electrophoresis. The cDNA Synthesis Kit (Illumina Inc., San Diego, CA, USA) was employed following the manufacturer's instructions to prepare cDNA for library construction. Illumina deep sequencing was carried out by Sangon Biotech Co., Ltd (Shanghai, China). Alignments were conducted using the tool package SOAP2 developed for short oligo nucleotide analysis, permitting up to 2 mismatches with reference sequences. Functional enrichment analyses were performed by R software (version 4.3.2) and drew by Bioinformatics online tool (www.bioinformatics.com.cn) to identify which DEGs were significantly enriched in metabolic pathways.

2.12. Immunofluorescence

Fix the mouse liver with 4% paraformaldehyde, followed by tissue dehydration and paraffin embedding. After sectioning, perform antigen retrieval. Subsequently, block the sections with 10% goat serum and incubate with anti-FOXO4 antibody. After washing, incubate with a fluorescently labeled secondary antibody at room temperature for 1 h, protected from light. After another wash, stain the nuclei with DAPI. Finally, mount the sections with an anti-fade mounting medium and observe the results under a fluorescence microscope.

2.13. Western blotting

The quality and quantity of total RNA were assessed by an UltrasecTM 2100 pro UV/Visible Spectrophotometer (Amersham Biosciences, Uppsala, Sweden) and agarose gel electrophoresis. The cDNA Synthesis Kit (Illumina Inc., San Diego, CA, USA) was employed following the manufacturer's instructions to prepare cDNA for library construction. Illumina deep sequencing was carried out by Sangon Biotech Co., Ltd (Shanghai, China). Alignments were conducted using the tool package SOAP2 developed for short oligo nucleotide analysis,

permitting up to 2 mismatches with reference sequences. The expression level of ERK, P-ERK, Foxo4, and HNF4a was represented by the optical density (OD) ratio to β -actin.

2.14. Statistical analysis

Results were presented as means $\pm$ standard deviation (S.D.) of the indicated number of independent experiments. In this study, we initially hypothesized whether the data satisfied the assumptions of normal distribution and homogeneity of variance. If these assumptions were not met, non-parametric test methods would be considered for further analysis. For data that conformed to normal distribution and homogeneity of variance, one-way analysis of variance (One – Way ANOVA) was used to evaluate the statistical significance. Statistical analysis was conducted using SPSS 20.0 software (SPSS Inc., Chicago, IL). A P value less than 0.05 was considered statistically significant.

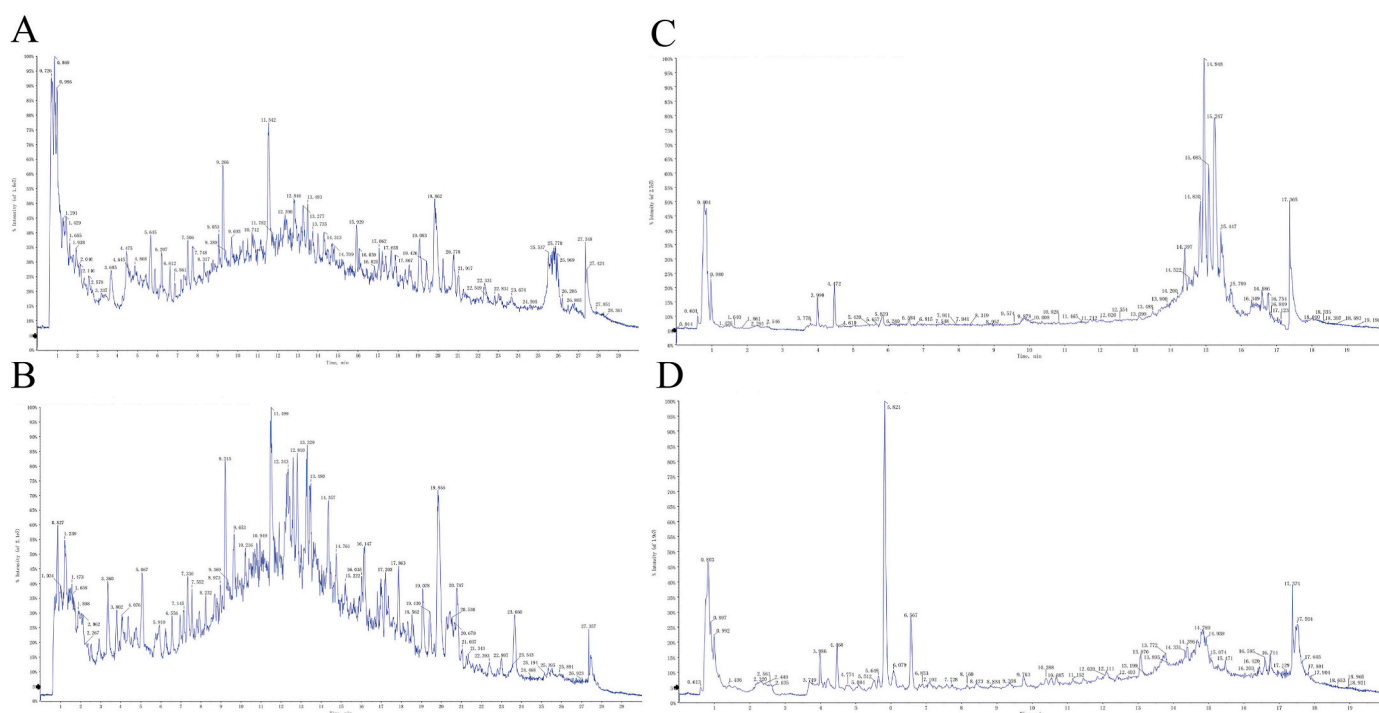
3. Result

3.1. Compounds characterization of HGJDD in decoction and drug serum

Fig. 1 illustrates the base peaks obtained from the HGJDD analysis in both positive and negative ion modes for the components of the decoction and drug serum. 127 compounds were identified in the decoction, while 106 compounds were characterized in the drug serum through LC-MS analysis, and there are 23 identical chemical components in the decoction and the serum, which may be caused by drug absorption and metabolism. Comprehensive details regarding these compounds can be found in [Supplementary Tables 1 and 2](#). For subsequent investigations, the TCMSP database was employed to predict the potential targets associated with these compounds.

3.2. Cytotoxicity of HGJDD serum

The CCK-8 assay was utilized to evaluate the viability of HepG2.2.15 cells when exposed to different concentrations of drug serum, aiming to



determine appropriate treatment concentrations for subsequent experiments on HBV-infected cell lines. The experimental design included six groups: normal group, control group, and groups treated with 10%, 20%, 40%, and 80% drug serum. The results indicated no significant cytotoxicity to HepG2.2.15 cells at the different concentrations of drug serum when compared to the control group over the periods of 3, 6, and 9 days (refer to Fig. 2A–C). Notably, the viability of cells in these groups was higher than that observed in the normal group (MEM + 10% FBS). Consequently, three concentrations—20%, 40%, and 80% drug serum—were selected for further experimentation.

3.3. HGJDD serum decrease the level of HBsAg, HBeAg, and HBV DNA in vitro

Following the administration of drug serum (comprising 20%, 40%, and 80% drug serum) to HepG2.2.15 cells over periods of 3, 6, and 9 days, a reduction in both extracellular and intracellular HBsAg and HBeAg was observed. This reduction exhibited both dose-dependent and time-dependent characteristics (refer to Fig. 2D, E, G, H). When compared to the control group, the serum containing the drug demonstrated a statistically significant inhibitory effect on extracellular HBsAg and HBeAg ($P < 0.05$). Furthermore, the inhibitory effects on both extracellular and intracellular DNA were also found to be dose-dependent and time-dependent (refer to Fig. 2F and I). Notably, the degree to which EVT treatment diminished HBV DNA and the reduction of the two types of HBV antigens differed from the effects observed with HGJDD serum.

3.4. HGJDD decrease the level of HBsAg, HBeAg, and HBV DNA in vivo

In the subsequent analysis, we evaluated levels of HBsAg, HBeAg, and HBV DNA in mice serum. Regarding serum HBsAg levels, both the high-dose HGJDD and EVT groups exhibited a gradual decrease post-administration. Conversely, the medium and low doses of HGJDD resulted in a gradual increase in HBsAg levels. Notably, after six weeks of treatment, the medium-dose HGJDD remained lower than that of the control group ($P < 0.05$). Although the low dose of HGJDD was also lower than the control group, the difference was not statistically significant. After six weeks, no significant difference was observed between the high-dose HGJDD group and the positive drug group (EVT), while the other groups demonstrated statistically significant differences in comparison to the positive drug group ($P < 0.05$) (refer to Fig. 3A). In terms of serum HBeAg, the high-dose HGJDD and positive drug groups showed a gradual decrease relative to the control group, while the medium and low doses of HGJDD exhibited a gradual increase. After six weeks, the medium dose of HGJDD remained lower than the control group ($P < 0.05$), and although the low dose was lower than the control group, this difference was not statistically significant. Furthermore, there were statistically significant differences among the various doses of HGJDD when compared to the positive drug group ($P < 0.05$) (refer to Fig. 3B). For serum HBV DNA, a statistically significant difference was noted between the high-dose HGJDD and control groups ($P < 0.05$). However, no significant differences were found between the high-dose or medium-dose HGJDD groups and the positive drug group, while the other groups exhibited statistical differences ($P < 0.05$) (refer to Fig. 3C). In terms of liver HBsAg levels, both the high-dose and medium-

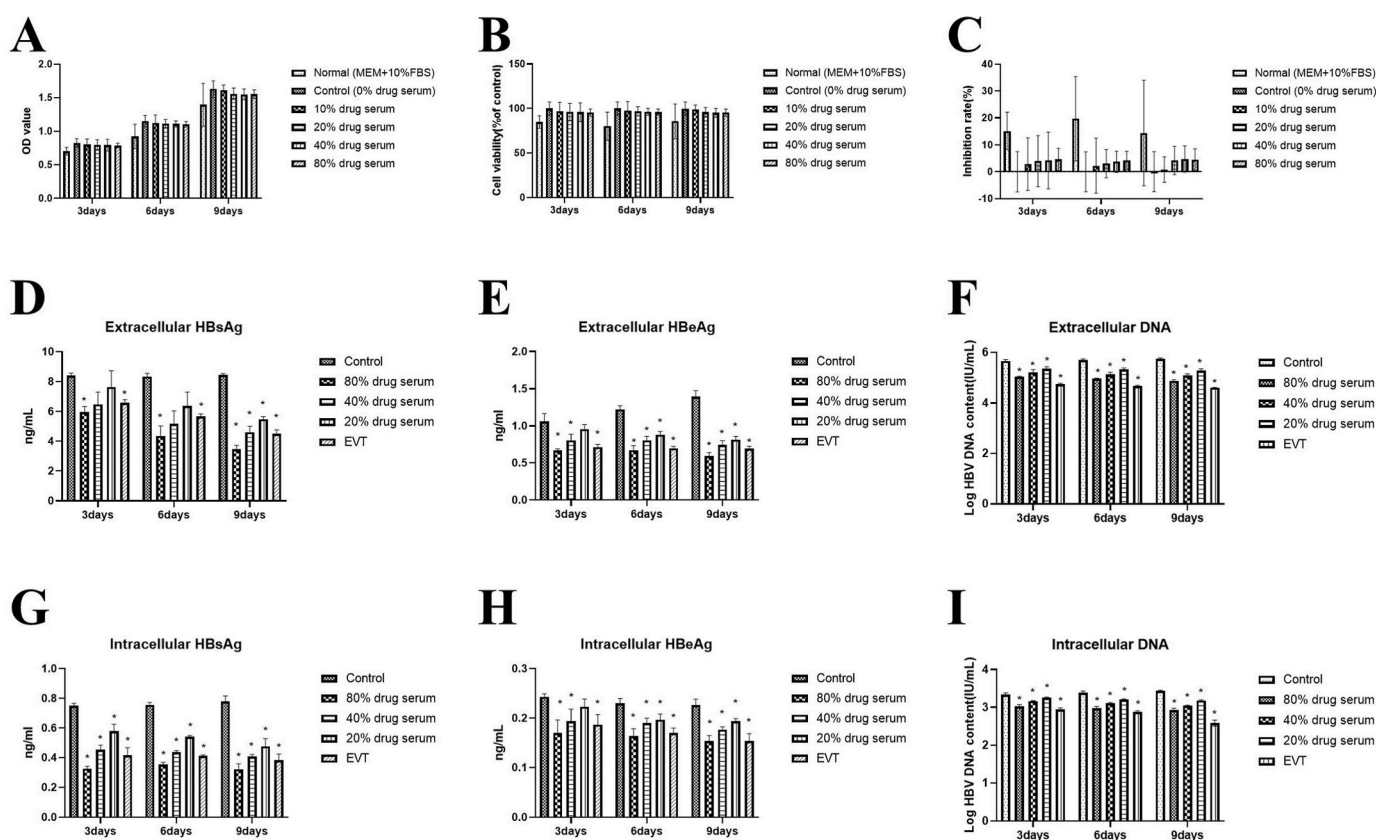


Fig. 2. Effects of different concentrations of drug-serum on HBsAg, HBeAg and DNA for Hepg2.2.15 cells. A. Changes of cell od value under different concentrations of drug-serum. B. Cell activity under different concentrations of drug-serum. C. Inhibitory effect on cells under different concentrations of drug-serum. D. Changes of HBsAg in cell supernatant under different concentrations of drug-serum. E. Changes of HBeAg in cell supernatant under different concentrations of drug-serum. F. Changes of HBV DNA in cell supernatant under different concentrations of drug-serum. G. Changes of intracellular HBsAg under different concentrations of drug-serum. H. Changes of intracellular HBeAg under different concentrations of drug-serum. I. Changes of intracellular HBV DNA under different concentrations of drug-serum.

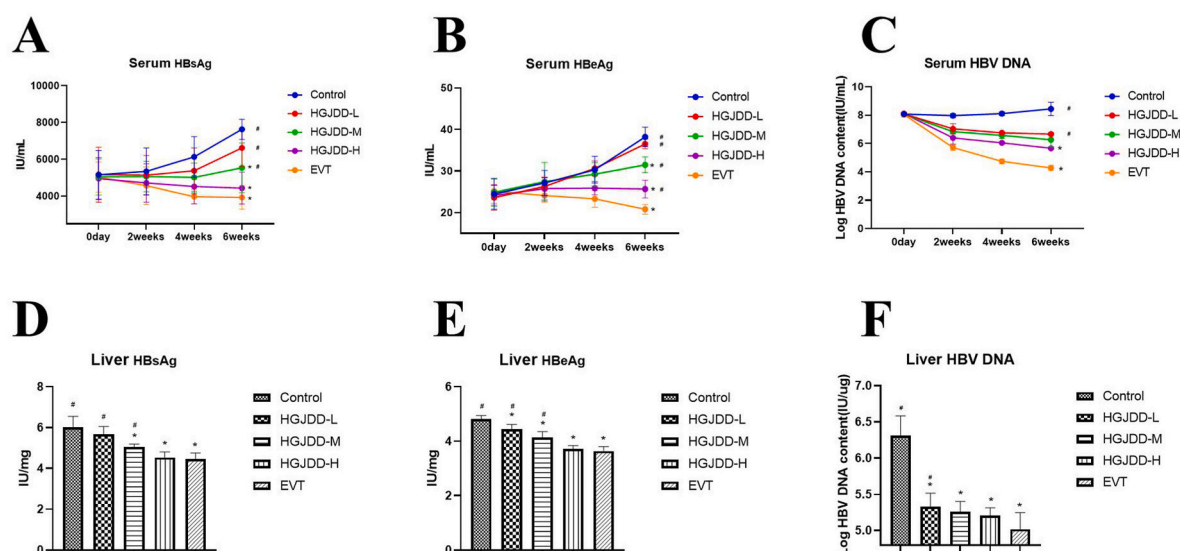


Fig. 3. Effects of different concentrations of HGJDD on HBsAg, HBeAg and DNA *in vivo*. A. Changes of HBsAg in serum. B. Changes of HBeAg in serum. C. Changes of HBV DNA in serum. D. Changes of HBsAg in liver. E. Changes of HBeAg in liver. F. Changes of HBV DNA in liver.

dose HGJDD groups exhibited statistically significant disparities compared with the control group ($P < 0.05$). Moreover, significant differences were noted among the control group, low-dose, and medium-dose HGJDD groups when contrasted with the positive drug group ($P < 0.05$). Regarding liver HBeAg, all groups showed statistically significant differences in relation to the control group ($P < 0.05$), except for the high-dose HGJDD group which also presented significant differences when compared to the positive drug group ($P < 0.05$). Finally, liver HBV DNA levels were significantly decreased in all groups ($P < 0.05$), and there were notable statistical differences between the positive drug group and both the control group and the low-dose HGJDD group ($P < 0.05$) (refer to Fig. 3D–F).

3.5. Network pharmaceutical analysis of drug decoctions and serum containing drugs

Utilizing the TCMSP database, we identified a total of 433 potential targets associated with the decoction (refer to Supplementary Table 3) and 245 potential targets related to the drug serum (see Supplementary Table 4). Additionally, an analysis of the GeneCards database and OMIM revealed 13,784 disease targets pertinent to HBV (as detailed in Supplementary Table 5). Following the removal of overlapping targets, we identified 268 unique genes as potential key targets for HBV treatment via decoction (Supplementary Table 6) and 165 unique genes as potential key targets for HBV treatment via drug serum (Supplementary Table 7). Building upon these common potential targets, we performed a bioinformatics analysis using R language software version 4.2.2. The analysis revealed the top ten signaling pathways exhibiting the most significant differences, as illustrated in Fig. 4A and B, with further details available in Supplementary Tables 8 and 9

3.6. Based on transcriptome analysis and intersection with network drug analysis

Utilizing differentially expressed genes, we performed a functional enrichment analysis and identified the ten signaling pathways exhibiting the most significant statistical differences (refer to Fig. 4C). Subsequently, we examined the overlap between the network pharmacology analysis of pertinent signaling pathways and the transcriptome analysis of associated pathways to identify shared signaling pathways. Our investigation revealed that the FOXO signaling pathway is a common pathway present at the intersection of the three analyses (refer to

Fig. 4D).

3.7. Immunofluorescence assay

The expression of FOXO4 in the liver was assessed using an immunofluorescence assay. Following a six-week treatment period, the findings indicated that FOXO4 expression exhibited varying degrees of increase in the groups receiving Chinese medicine, with the most pronounced elevation observed in the high-dose HGJDD group. In contrast, FOXO4 expression was minimal in both the control group and the positive drug group (refer to Fig. 5).

3.8. Western blotting

Following a six-week administration of HGJDD, liver samples from the animals were analyzed to identify associated protein alterations, as illustrated in Fig. 6 (* indicates $0.01 \leq P < 0.05$; ** denotes $0.001 \leq P < 0.01$; *** signifies $P < 0.001$). The expression levels of the FOXO4 protein in all three experimental groups were found to be significantly elevated compared to the control group. Similarly, the expression of the ERK protein was also significantly higher in the three groups relative to the control group. It is worth noting that the levels of p-ERK protein in the high-dose and medium-dose groups were markedly increased in comparison with the control group. However, no statistically significant difference was found between the low-dose group and the control group. Moreover, the expression of HNF4 α protein in both the high-dose and medium-dose groups was significantly higher than that in the control group. On the other hand, no significant difference was detected between the low-dose group and the control group. Additionally, no significant changes in protein were observed between the positive drug group and the control group.

4. Discussion

In this investigation, we utilized the widely recognized herbal formulation, HuaganJiedu Decoction, which has been clinically validated for its therapeutic efficacy in the treatment of hepatitis B. Initially, we employed LC-MS to analyze the composition of the decoction and the serum containing the active drug components, thereby identifying the chemical constituents that may contribute to its therapeutic effects. Following this, we conducted experiments using serum containing the drug on the HepG2.2.15 cell line, one well-established cell line for

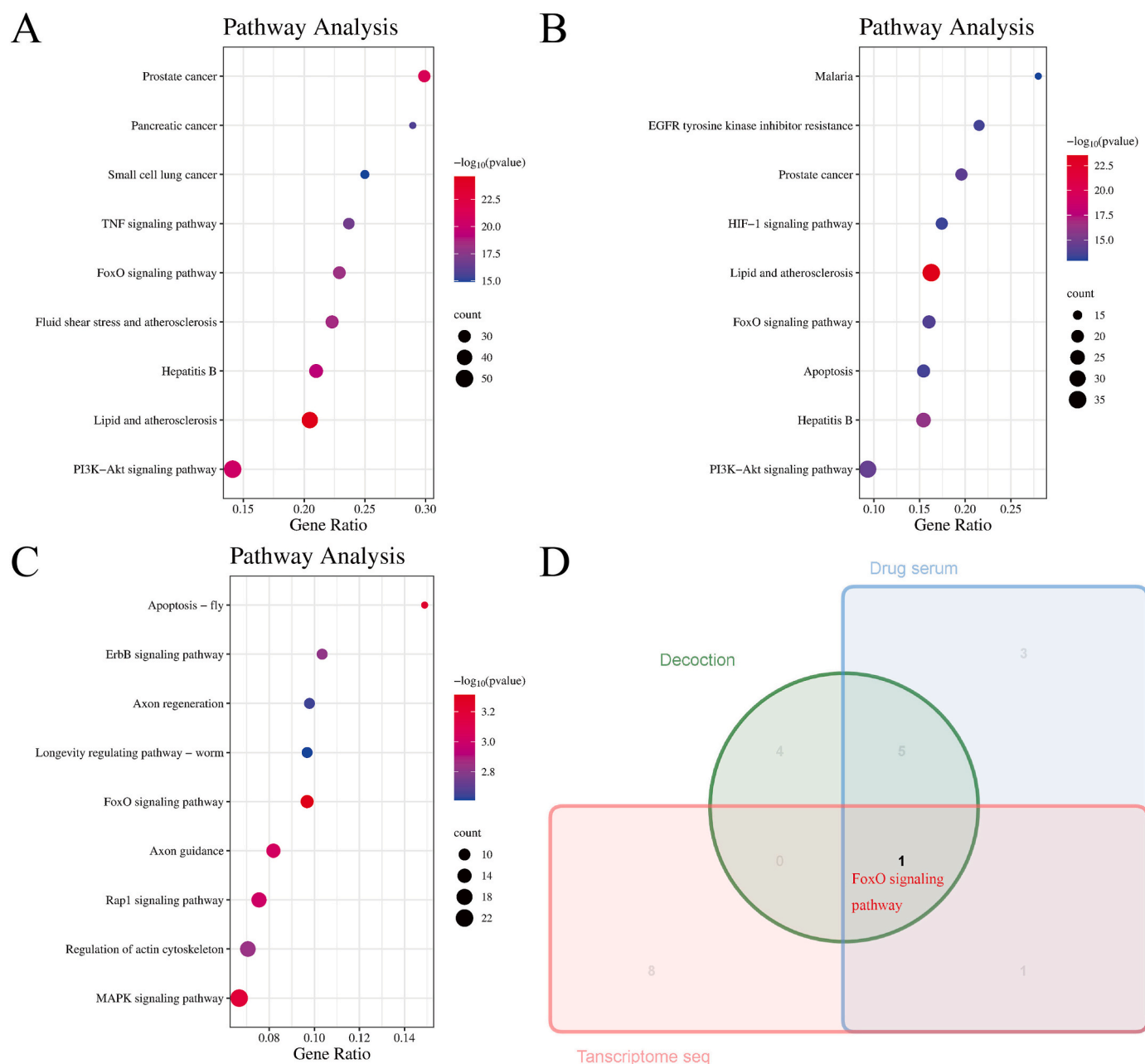


Fig. 4. Biological information analysis of the mechanism of HGJDD. A. Top ten signal pathways with statistical difference based on network pharmacological analysis of the therapeutic effect of decoction composition on hepatitis B. B. Top 10 signaling pathways with statistical difference based on network pharmacological analysis of the therapeutic effect of drug-containing serum components on hepatitis B. C. The top 10 signaling pathways with the most statistically significant differences were analyzed based on the transcriptome. D. The first three groups of the top ten signal paths intersect.

assessing pharmacological agents in anti-HBV (Han et al., 2011). Cell viability was evaluated through the CCK-8 assay across various concentrations of drug-containing serum to determine the suitable treatment concentrations for follow-up experiments involving HBV-infected cell cultures. Comparative analysis among six groups—normal control, control, 10% drug serum, 20% drug serum, 40% drug serum, and 80% drug serum—indicated that the drug-containing serum significantly enhanced cell proliferation compared to the standard complete medium. However, there were no statistically significant differences in cell proliferation when comparing different concentrations of drug-containing serum at days 3, 6, and 9. Consequently, subsequent experiments concentrated on drug-containing serum concentrations of 20%, 40%, and 80%. Hepatitis B virus (HBV), which is a partially double-stranded DNA virus, secretes various serological markers such as HBsAg and

HBeAg, which were detected in the HepG2.2.15 cell line, alongside the synthesis of Dane particles. The inhibition of HBsAg, HBeAg and HBV DNA, serve as reliable indicators for the screening of anti-HBV agents. Following treatment with 20%, 40%, and 80% drug-containing serum over periods of 3, 6, and 9 days, the findings revealed that HuaganJiedu Decoction significantly diminished the secretion of HBsAg and HBeAg and suppressed DNA levels across all three concentrations. Furthermore, its inhibitory effects on both intra- and extracellular HBsAg, HBeAg, and HBV DNA were dose- and time-dependent. Notably, the efficacy of the 80% drug-containing serum in inhibiting HBsAg and HBeAg secretion surpassed that of the antiviral agent EVT, although it was less effective than EVT in inhibiting HBV DNA levels. This suggests that other components within the herbal formulation may inhibit the secretion and synthesis of HBsAg and HBeAg. Previous research has also indicated

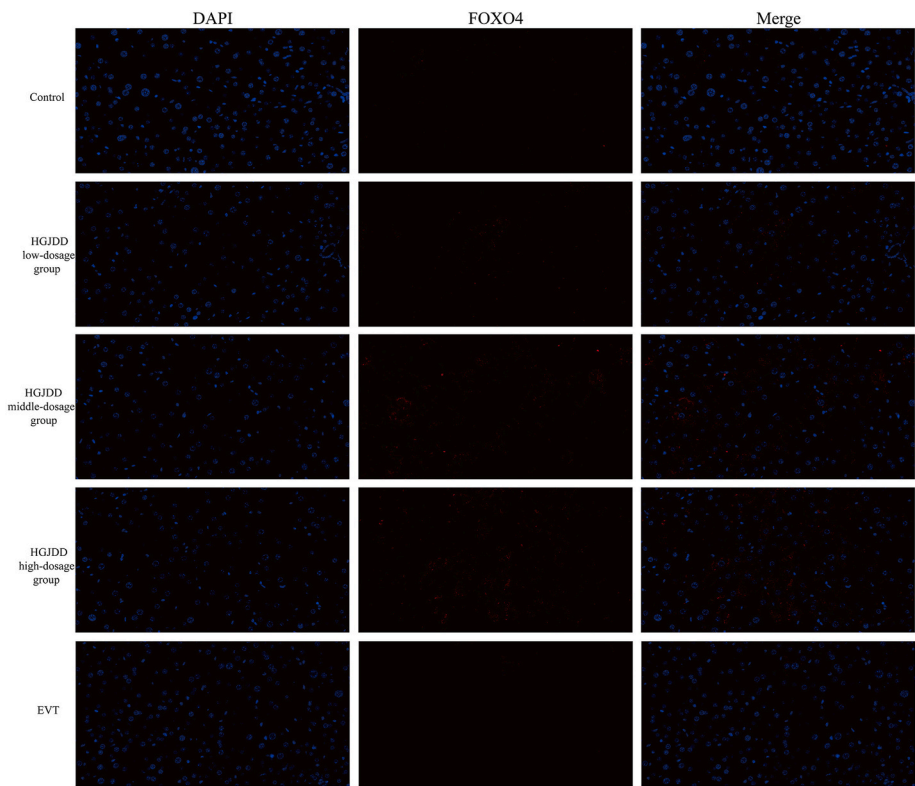


Fig. 5. Effects of each group on FOXO4D protein expression by immunofluorescence.

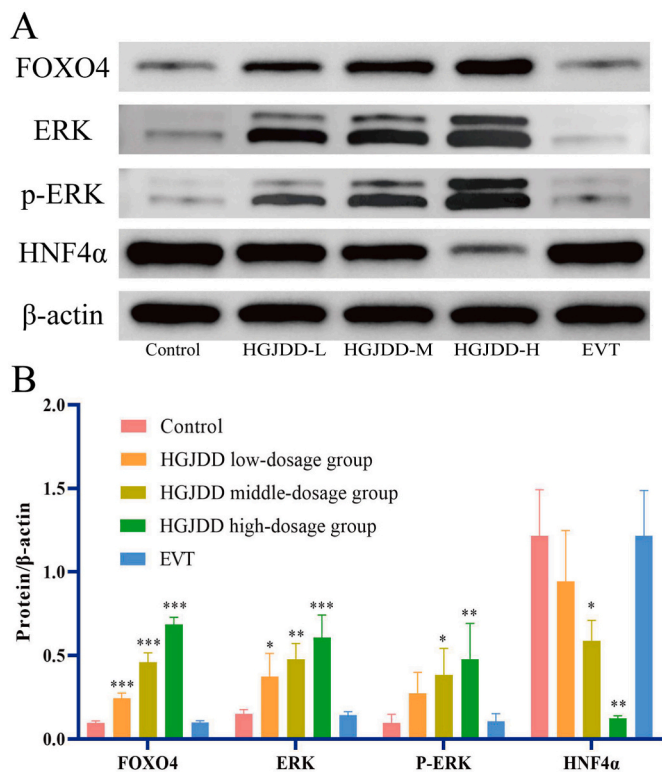


Fig. 6. The expression level of related mechanism protein. A. Western blots showing the levels of FOXO4, ERK, p- ERK, HNF4α; B. showed expression of each protein normalized by β-actin according to grey value.

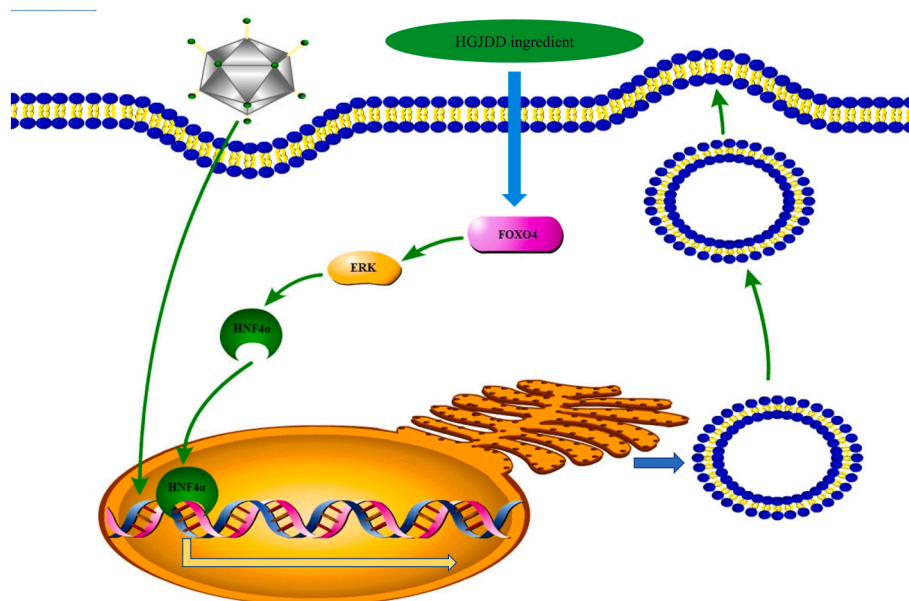
significant reductions in both intra- and extracellular HBV DNA levels following drug treatment, despite no notable changes in HBsAg and HBeAg mRNA expression levels, ultimately leading to decreased levels of HBsAg and HBeAg (Wu et al., 2016).

Subsequently, we assessed the therapeutic effects of various dosages of HGJDD on HBV transgenic mice (Tg mice). As of the sixth week of treatment, when compared with the control group, significant decreases in HBsAg were observed in the high-dose group, medium-dose group, and positive drug group; however, the medium-dose group demonstrated less pronounced reductions relative to the positive drug group. Furthermore, while the medium-dose group displayed lower levels than the control group, an increasing trend persisted, underscoring the critical importance of achieving an adequate concentration of traditional Chinese medicine to elicit an inhibitory effect. Similar trends were observed for HBeAg, with significant reductions in the high-dose HGJDD, medium-dose HGJDD, and positive drug groups compared to the control group at the sixth week of treatment, and all groups exhibited lower efficacy than the positive drug group. Notably, significant reductions in HBV DNA were recorded in the high-dose HGJDD and positive drug group compared with the control group by the sixth week of treatment. The serum levels of these indicators suggest that the dosage of HGJDD plays a crucial role in determining its efficacy. While lower doses may partially inhibit HBsAg and HBeAg, the secretion levels in the animal model continued to rise. Given the transgenic mice's tolerance to the human hepatitis B virus, it is evident that although varying dosages resulted in different degrees of reduction in HBV DNA, the effects on HBsAg and HBeAg secretion were not as substantial. This discrepancy may indicate a reduction in the release of viral particles and genome replication within liver; however, HBsAg and HBeAg continued to be released into the bloodstream without being cleared by the mice's immune system. The inhibitory effects observed on these indicators in this mouse model are consistent with findings from previous research (Li et al., 2018a; Zhang et al., 2024).

We undertook a qualitative analysis of the herbal constituents present in the decoction and serum containing the drug. The specific

Emerging evidence indicates that Forkhead box O (FoxO) proteins may regulate HBV infection by activating the its core promoter, which is integral to HBV transcription and replication (Shlomai and Shaul, 2009; Wang and Tian, 2017). *In vitro* and *in vivo*, it had demonstrated that FOXO4 exerts inhibitory effects on HBV transcription and replication. Further research has shown that the FOXO 4-mediated inhibition of HBV core promoter activity is linked to the downregulation of hepatocyte nuclear factor 4 alpha (HNF4 α) via the extracellular signal-regulated kinase (ERK) signaling pathway (Li et al., 2019). This finding is consistent with our bioinformatics analysis, which suggests that traditional Chinese medicine (TCM) may primarily function through this pathway. Additionally, our transcriptomic analysis highlighted the involvement of the mitogen-activated protein kinase (MAPK) signaling pathway, and it is vital for FoxO activity. The MAPK signaling pathway has been reported that could regulate HNF4 α expression, and there is evidence to suggest that FOXO proteins are very important in the modulation of this pathway (de Boussac et al., 2010; Hatzis et al., 2006; Li et al., 2019). Moreover, immunofluorescence studies revealed an increase in FOXO4 protein expression following the administration of TCM, with a correlation observed between FOXO4 levels and the concentration of the medicine, aligning with our transcriptomic findings. Subsequent Western blot analyses demonstrated elevated expression levels of FOXO4, ERK, phosphorylated ERK (P-ERK), and decreased

In conclusion, although vaccines are widely available, HBV infection remains the most common chronic viral disease globally. Although NAs have significantly improved treatment outcomes, certain patients still experience inadequate responses and challenges in deriving benefits from these therapies. Traditional Chinese medicine (TCM) presents a variety of herbal formulations for the management of hepatitis B and has amassed considerable clinical experience in this area. This study focused on HGJDD, a representative TCM formula, and demonstrated its therapeutic efficacy against HBV at both cellular and animal model levels, resulting in decreased HBsAg, HBeAg, and HBV DNA. Utilizing network



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pharmacology in conjunction with bioinformatics analysis and protein-level validation, the research identified a significant relationship between the therapeutic effects of HGJDD and the FOXO4/ERK/HNF4 α signaling pathway. The elucidation of this mechanism offers a scientific basis for employing HGJDD in the treatment of HBV.

CRedit authorship contribution statement

Hongxuan Tong: Writing – review & editing, Writing – original draft, Supervision, Investigation, Data curation, Conceptualization. **Jiale Zhang:** Data curation. **Lijie Jiang:** Methodology. **Rendong Qu:** Methodology. **Tao Lu:** Writing – review & editing, Software. **Jingqing Hu:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2024.119238>.

Data availability

Data will be made available on request.

References

- Chen, F., 2006. Clinical effect evaluation of Huguang detoxification decoction combined with interferon in the treatment of hepatitis B. *West. China J. Pharm. Sci.* 21, 621–622. <https://doi.org/10.13375/j.cnki.wcjps.2006.06.053>.
- Chen, V.L., Yeh, M.L., Le, A.K., Jun, M., Saeed, W.K., Yang, J.D., Huang, C.F., Lee, H.Y., Tsai, P.C., Lee, M.H., Giama, N., Kim, N.G., Nguyen, P.P., Dang, H., Ali, H.A., Zhang, N., Huang, J.F., Dai, C.Y., Chuang, W.L., Roberts, L.R., Jun, D.W., Lim, Y.S., Yu, M.L., Nguyen, M.H., 2018. Anti-viral therapy is associated with improved survival but is underutilised in patients with hepatitis B virus-related hepatocellular carcinoma: real-world east and west experience. *Aliment. Pharmacol. Ther.* 48, 44–54. <https://doi.org/10.1111/apt.14801>.
- Chen, Y., Zhu, J., 2013. Anti-HBV effect of individual traditional Chinese herbal medicine *in vitro* and *in vivo*: an analytic review. *J. Viral Hepat.* 20, 445–452. <https://doi.org/10.1111/jvh.12112>.
- Cui, X., Wang, Y., Kokudo, N., Fang, D., Tang, W., 2010. Traditional Chinese medicine and related active compounds against hepatitis B virus infection. *Biosci. Trends* 4, 39–47.
- Dai, X.Q., Cai, W.T., Wu, X., Chen, Y., Han, F.M., 2017. Protocatechuic acid inhibits hepatitis B virus replication by activating ERK1/2 pathway and down-regulating HNF4 α and HNF1 α *in vitro*. *Life Sci.* 180, 68–74. <https://doi.org/10.1016/j.lfs.2017.05.015>.
- de Bousac, H., Ratajewski, M., Sachrajda, I., Köblös, G., Tordai, A., Pulaski, L., Buday, L., Váradi, A., Arányi, T., 2010. The ERK1/2-hepatocyte nuclear factor 4 α axis regulates human ABCG6 gene expression in hepatocytes. *J. Biol. Chem.* 285, 22800–22808. <https://doi.org/10.1074/jbc.M110.105593>.
- de Martel, C., Georges, D., Bray, F., Ferlay, J., Clifford, G.M., 2020. Global burden of cancer attributable to infections in 2018: a worldwide incidence analysis. *Lancet Global Health* 8, e180–e190. [https://doi.org/10.1016/S2214-109X\(19\)30488-7](https://doi.org/10.1016/S2214-109X(19)30488-7).
- Fanning, G.C., Zoulim, F., Hou, J., Bertoletti, A., 2019. Therapeutic strategies for hepatitis B virus infection: towards a cure. *Nat. Rev. Drug Discov.* 18, 827–844. <https://doi.org/10.1038/s41573-019-0057-9>.
- Han, Q., Zhang, C., Zhang, J., Tian, Z., 2011. Reversal of hepatitis B virus-induced immune tolerance by an immunostimulatory 3p-HBx-siRNAs in a retinoic acid inducible gene I-dependent manner. *Hepatology* 54, 1179–1189. <https://doi.org/10.1002/hep.24505>.
- Hatzakis, A., Van Damme, P., Alcorn, K., Gore, C., Benazzouz, M., Berkane, S., Buti, M., Carballo, M., Cortes Martins, H., Deuffic-Burban, S., Dominguez, A., Donoghoe, M., Elzouki, A.N., Ben-Alaya Bouafif, N., Esmat, G., Esteban, R., Fabri, M., Fenton, K., Goldberg, D., Goulis, I., Hadjichristodoulou, C., Hatzigeorgiou, T., Hamouda, O., Hasurdjiev, S., Hughes, S., Kautz, A., Malik, M., Manolakopoulos, S., Maticic, M., Papatheodoridis, G., Peck, R., Peterle, A., Potamitis, G., Prati, D., Roudot-Thoraval, F., Reic, T., Sharara, A., Shennak, M., Shih, G., Shouval, D., Sočan, M., Thomas, H., Thursz, M., Tosti, M., Trépo, C., Vince, A., Vounou, E., Wiessing, L., Manns, M., 2013. The state of hepatitis B and C in the Mediterranean and Balkan countries: report from a summit conference. *J. Viral Hepat.* 20 (Suppl. 2), 1–20. <https://doi.org/10.1111/jvh.12120>.
- Hatzis, P., Kyrmizi, I., Talianidis, I., 2006. Mitogen-activated protein kinase-mediated disruption of enhancer-promoter communication inhibits hepatocyte nuclear factor 4 α expression. *Mol. Cell Biol.* 26, 7017–7029. <https://doi.org/10.1128/MCB.00297-06>.
- Huang, D.Q., Tran, A., Yeh, M.L., Yasuda, S., Tsai, P.C., Huang, C.F., Dai, C.Y., Ogawa, E., Ishigami, M., Ito, T., Kozuka, R., Enomoto, M., Suzuki, T., Yoshimaru, Y., Preda, C. M., Marin, R.I., Sandra, I., Tran, S., Quek, S.X.Z., Khine, H., Itokawa, N., Atsukawa, M., Uojima, H., Watanabe, T., Takahashi, H., Inoue, K., Maeda, M., Hoang, J.K., Trinh, L., Barnett, S., Cheung, R., Lim, S.G., Trinh, H.N., Chuang, W.L., Tanaka, Y., Toyoda, H., Yu, M.L., Nguyen, M.H., 2023. Antiviral therapy substantially reduces HCC risk in patients with chronic hepatitis B infection in the indeterminate phase. *Hepatology* 78, 1558–1568. <https://doi.org/10.1097/HEP.0000000000000459>.
- Huang, H., Tindall, D.J., 2007. Dynamic FoxO transcription factors. *J. Cell Sci.* 120, 2479–2487. <https://doi.org/10.1242/jcs.001222>.
- Kaixian, C., 2020. Academician chen kaixiantalks about the development of traditional Chinese medicine and global medicine. *World J. Tradit. Chin. Med.* 6, 1.
- Kudaravalli, S., Huang, D.Q., Yeh, M.L., Trinh, L., Tsai, P.C., Hsu, Y.C., Kam, L.Y., Nguyen, V.H., Ogawa, E., Lee, D.H., Ito, T., Watanabe, T., Enomoto, M., Preda, C.M., Ko, M.K.L., Wan-Hin Hui, R., Atsukawa, M., Suzuki, T., Marciano, S., Barreira, A., Do, S., Uojima, H., Takahashi, H., Quek, S.X.Z., Toe Wai Khine, H.H., Ishigami, M., Itokawa, N., Go, M.S., Kozuka, R., Marin, R.I., Sandra, I., Li, J., Zhang, J.Q., Wong, C., Yoshimaru, Y., Vo, D.K.H., Tseng, C.H., Lee, C.J., Inoue, K., Maeda, M., Hoang, J.K., Chau, A., Chuang, W.L., Dai, C.Y., Huang, J.F., Huang, C.F., Buti, M., Tanaka, Y., Gadano, A.C., Yuen, M.F., Cheung, R., Lim, S.G., Trinh, H.N., Toyoda, H., Yu, M.L., Nguyen, M.H., 2024. Sex and ethnic disparities in hepatitis B evaluation and treatment across the world. *J. Hepatol.* 81, 33–41. <https://doi.org/10.1016/j.jhep.2024.02.033>.
- Laupéze, B., Vassilev, V., Badur, S., 2023. A role for immune modulation in achieving functional cure for chronic hepatitis B among current changes in the landscape of new treatments. *Expert Rev. Gastroenterol. Hepatol.* 17, 1135–1147. <https://doi.org/10.1080/17474124.2023.2268503>.
- Lebossé, F., Testoni, B., Fresquet, J., Facchetti, F., Galmozzi, E., Fournier, M., Hervieu, V., Berthillon, P., Berby, F., Bordes, I., Durantal, D., Levrero, M., Lampertico, P., Zoulim, F., 2017. Intrahepatic innate immune response pathways are downregulated in untreated chronic hepatitis B. *J. Hepatol.* 66, 897–909. <https://doi.org/10.1016/j.jhep.2016.12.024>.
- Li, H., Sheng, C., Liu, H., Wang, S., Zhao, J., Yang, L., Jia, L., Li, P., Wang, L., Xie, J., Xu, D., Sun, Y., Qiu, S., Song, H., 2018a. Inhibition of HBV expression in HBV transgenic mice using AAV-delivered CRISPR-SaCas9. *Front. Immunol.* 9, 2080. <https://doi.org/10.1016/j.jhep.2016.12.024>.
- Li, J., Wang, X., Chen, Y., 2018b. Clinical study of Huangjiedu Decoction combined with Entecavir in the treatment of chronic viral hepatitis B. *Pract. Clin. Integr. J. Chin. West. Med.* 18, 73–74. <https://doi.org/10.13638/j.issn.1671-4040.2018.12.035>.
- Li, L., Li, Y., Xiong, Z., Shu, W., Yang, Y., Guo, Z., Gao, B., 2019. FoxO4 inhibits HBV core promoter activity through ERK-mediated downregulation of HNF4 α . *Antivir. Res.* 170, 104568. <https://doi.org/10.1016/j.antiviral.2019.104568>.
- Liu, C., Zhao, M., Cheng, R., 2010. Zhou Zhongying's experience in treating hepatitis B with Huangjiedu decoction. *Med. Res. Educ.* 27, 63–64.
- Lok, A.S., Zoulim, F., Dusheiko, G., Ghany, M.G., 2017. Hepatitis B cure: from discovery to regulatory approval. *Hepatology* 66, 1296–1313. <https://doi.org/10.1002/hep.29323>.
- Nguyen, V.H., Le, A.K., Trinh, H.N., Chung, M., Johnson, T., Wong, C., Wong, C., Zhang, J., Li, J., Levitt, B.S., Nguyen, H.A., Nguyen, K.K., Henry, L., Cheung, R., Nguyen, M.H., 2019. Poor adherence to guidelines for treatment of chronic hepatitis B virus infection at primary care and referral practices. *Clin. Gastroenterol. Hepatol.* 17, 957–967.e957. <https://doi.org/10.1016/j.cgh.2018.10.012>.
- Orea-Soufi, A., Paik, J., Bragança, J., Donlon, T.A., Willcox, B.J., Link, W., 2022. FOXO transcription factors as therapeutic targets in human diseases. *Trends Pharmacol. Sci.* 43, 1070–1084. <https://doi.org/10.1016/j.tips.2022.09.010>.
- Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y., Xu, X., Li, Y., Wang, Y., Yang, L., 2014. TCMSP: a database of systems pharmacology for drug discovery from herbal medicines. *J. Cheminf.* 6, 13. <https://doi.org/10.1186/1758-2946-6-13>.
- Safran, M., Dalah, I., Alexander, J., Rosen, N., Iny Stein, T., Shmoish, M., Nativ, N., Bahir, I., Doniger, T., Krug, H., Sirota-Madi, A., Olender, T., Golan, Y., Stelzer, G., Harel, A., Lancet, D., 2010. GeneCards Version 3: the human gene integrator. *Database* 2010, baq020. <https://doi.org/10.1093/database/baq020>.
- Shlomai, A., Shaul, Y., 2009. The metabolic activator FOXO1 binds hepatitis B virus DNA and activates its transcription. *Biochem. Biophys. Res. Commun.* 381, 544–548. <https://doi.org/10.1016/j.bbrc.2009.02.078>.
- Sun, G., Mao, C., 2019. Clinical effect evaluation of huan detoxification decoction combined with interferon in the treatment. *J. Clin. Med.* 6, 2–3.
- Terrault, N.A., Lok, A.S.F., McMahon, B.J., Chang, K.M., Hwang, J.P., Jonas, M.M., Brown Jr., R.S., Bzowej, N.H., Wong, J.B., 2018. Update on prevention, diagnosis,

- and treatment of chronic hepatitis B: AASLD 2018 hepatitis B guidance. Clin. Liver Dis. 12, 33–34. <https://doi.org/10.1002/cld.728>.
- Trépo, C., Chan, H.L., Lok, A., 2014. Hepatitis B virus infection. Lancet 384, 2053–2063. [https://doi.org/10.1016/S0140-6736\(14\)60220-8](https://doi.org/10.1016/S0140-6736(14)60220-8).
- Wang, T., Wang, X., Zhuo, Y., Si, C., Yang, L., Meng, L., Zhu, B., 2020. Antiviral activity of a polysaccharide from *Radix Isatidis* (*Isatis indigotica* Fortune) against hepatitis B virus (HBV) *in vitro* via activation of JAK/STAT signal pathway. J. Ethnopharmacol. 257, 112782. <https://doi.org/10.1016/j.jep.2020.112782>.
- Wang, Y., Tian, H., 2017. miR-101 suppresses HBV replication and expression by targeting FOXO1 in hepatoma carcinoma cell lines. Biochem. Biophys. Res. Commun. 487, 167–172. <https://doi.org/10.1016/j.bbrc.2017.03.171>.
- Webb, A.E., Kundaje, A., Brunet, A., 2016. Characterization of the direct targets of FOXO transcription factors throughout evolution. Aging Cell 15, 673–685. <https://doi.org/10.1111/acer.12479>.
- Wu, L., Wang, W., Zhang, X., Zhao, X., Yu, G., 2016. Anti-HBV activity and mechanism of marine-derived polyguluronate sulfate (PGS) *in vitro*. Carbohydr. Polym. 143, 139–148. <https://doi.org/10.1016/j.carbpol.2016.01.065>.
- Xu, H., Zhang, Y., Wang, P., Zhang, J., Chen, H., Zhang, L., Du, X., Zhao, C., Wu, D., Liu, F., Yang, H., Liu, C., 2021. A comprehensive review of integrative pharmacology-based investigation: a paradigm shift in traditional Chinese medicine. Acta Pharm. Sin. B 11, 1379–1399. <https://doi.org/10.1016/j.apsb.2021.03.024>.
- Yin, X., Li, J., Hao, Z., Ding, R., Qiao, Y., 2022. A systematic study of traditional Chinese medicine treating hepatitis B virus-related hepatocellular carcinoma based on target-driven reverse network pharmacology. Front. Cell. Infect. Microbiol. 12, 964469. <https://doi.org/10.3389/fcimb.2022.964469>.
- Yuan, F., Jun, S., Oian, G., Yiqun, Z., Fang, X., Hailin, T., Yuiie, G., Yueqiu, G., 2022. Study on clinical efficacy of Huagan Jiedu decoction on patients with chronic hepatitis B and effects of peripheral blood T lymphocytes, IL-6 and GP73. J. Basic Chin. Med. 28, 1126–1129+1178. <https://doi.org/10.19945/j.cnki.issn>.
- Yuen, M.F., Chen, D.S., Dusheiko, G.M., Janssen, H.L.A., Lau, D.T.Y., Locarnini, S.A., Peters, M.G., Lai, C.L., 2018. Hepatitis B virus infection. Nat. Rev. Dis. Primers. 4, 18035. <https://doi.org/10.1038/nrdp.2018.35>.
- Zhang, C., Chu, L., Li, K., Yan, M., 2024. Release mechanism and pharmacodynamics of entecavir micro spheres. Pak. J. Pharm. Sci. 37, 107–113.
- Zhang, L., Wang, G., Hou, W., Li, P., Dulin, A., Bonkovsky, H.L., 2010. Contemporary clinical research of traditional Chinese medicines for chronic hepatitis B in China: an analytical review. Hepatology 51, 690–698. <https://doi.org/10.1002/hep.23384>.
- Zhou, C., Wei, J., Yu, P., Yang, J., Liu, T., Jia, R., Wang, S., Sun, P., Yang, L., Xiao, H., 2023. Convergent application of traditional Chinese medicine and gut microbiota in ameliorate of cirrhosis: a data mining and Mendelian randomization study. Front. Cell. Infect. Microbiol. 13, 1273031. <https://doi.org/10.3389/fcimb.2023.1273031>.